

MOYI TIAN

Applied Mathematics, 526 UCB, Boulder, CO, 80309, United States

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🌐 <https://moyi-tian.github.io/moyi-tian>

ACADEMIC APPOINTMENTS

University of Colorado Boulder, Boulder, CO

August 2024 - Present

Postdoctoral Associate

PI: Nancy Rodriguez (co-PI: David Bortz)

EDUCATION

Brown University, Providence, RI

September 2019 - May 2024

Ph.D in Applied Mathematics

Thesis Advisor: Dr. Björn Sandstede, Division of Applied Mathematics

Thesis: Patterns in Network Dynamics

MS, Applied Mathematics

May 2021

Dickinson College, Carlisle, PA

September 2015 - May 2019

B.S. in Mathematics & Physics

Phi Beta Kappa Honors, Honors in Mathematics, *Summa Cum Laude*

PUBLICATIONS

Published Articles

- 2023 **M Tian** and P Moriano, *Robustness of community structure under edge addition*, Phys. Rev. E **108** (2023), 054302
- 2021 **M Tian**, JJ Bramburger, and B Sandstede, *Snaking Bifurcations of Localized Patterns on Ring Lattices*, IMA Journal of Applied Mathematics (2021), hxab023
- 2019 H Mhiri, **M Tian**, E Wynne, S Jones, A Mareno, and LQ English, *An Experimental Survey of Chaos and Symmetry Breaking in Coupled and Driven Logistic Maps*, European Journal of Physics **40** (2019), no. 6, 065802

SELECTED AWARDS, HONORS AND MEMBERSHIPS

Awards

- 2025 **Dynamics Days Travel Awards**, Dynamics Days US 2025
Travel support to attend Dynamics Days US 2025 conference
- 2024 **SIAM Student Chapter Certificate of Recognition**, Society for Industrial and Applied Mathematics
In recognition of outstanding efforts and accomplishments on behalf of the SIAM Chapter at Brown University
- 2023 **SIAM Student Travel Awards**, Society for Industrial and Applied Mathematics
Support students to attend and present at SIAM conferences

- 2019 **James Fowler Rusling Prize**, Dickinson College
Presented to a member of the senior class whose scholarly achievements have been judged most superior by the All-College Committee on Academic Program and Standards
- The Lance E. Kohlhaas Memorial Prize in Mathematics**, Dickinson College
Awarded to a graduating mathematics major who has demonstrated excellence in that field and shows promise in an actuarial or mathematics career
- 2018 **The Caroline Hatton Clark Mathematics Scholarship**, Dickinson College
Awarded for outstanding achievement in mathematics
- 2017 **The Henry P. Cannon Memorial Prize**, Dickinson College
Awarded to a member of the sophomore class who excels in mathematics
- The Junior Class Sophister Prize**, Dickinson College
Awarded to the junior with the highest academic ranking at the start of the academic year
- 2016 **The John Patton Memorial Prize**, Dickinson College
Awarded to a rising sophomore for high scholastic standing

Honors

Phi Beta Kappa Honor Society
Pi Mu Epsilon National Honorary Mathematics Society
Sigma Pi Sigma National Physics Honor Society
Alpha Lambda Delta Honor Society

Memberships

2025 - Present Association for Women in Mathematics
2019 - Present Society for Industrial and Applied Mathematics
2019 - 2024 American Mathematical Society

PRESENTATIONS

Talks

- 2025 **ICERM Workshop on Patterns, Dynamics, and Data in Complex Systems**
Localized Patterns and Modeling Dynamics on Networks, Providence, RI, January, 2025
- 2024 **2024 Joint Mathematics Meetings (JMM 2024)**
Inferring Interaction Kernels for Stochastic Agent-Based Opinion Dynamics, San Francisco, CA, January, 2024
- 2023 **SIAM Conference on Applications of Dynamical Systems (DS23)**
Localized Patterns on Graphs, Portland, OR, May, 2023
- 2022 **SIAM Workshop on Network Science (NS22)** (virtual) - Lightning Talk
How Robust are Communities in Temporal Networks? A Comparative Analysis Using Community Detection Algorithms, September, 2022
- 2021 **Graduate Seminar**
Localized Patterns on Symmetric Coupled Rings - The Influence of Interaction Length on Pattern Formation, Brown University, Providence, RI, December, 2021
- 2021 **Brown / BU / UMass Dynamics & PDE Seminar**
Snaking Bifurcations of Localized Patterns, University of Massachusetts Amherst, Amherst, MA, November, 2021

Posters

- 2025 **SIAM Conference on Applications of Dynamical Systems (DS25)**
Modeling Coupled Online and Offline Dynamics of Protesting Activity on Networks, Denver, CO, May, 2025
- 2025 **Dynamics Days US 2025** - Flash Talk & Poster
Localized Patterns on Ring Lattices, Denver, CO, January, 2025
- 2024 **AWM Workshop at SIAM Annual Meeting 2024**
Efficiently Learning Models of Networks, Spokane, WA, July, 2024
- 2024 **2024 Dynamics Days US**
Efficient Learning of Models for Temporal Networks, Davis, CA, January, 2024
- 2023 **Dynamics Days US 2023** (virtual)
Community Robustness in Temporal Networks under Edge Addition, January, 2023
- 2021 **SIAM Conference on Applications of Dynamical Systems (DS21)** (virtual)
Snaking Bifurcations of Localized Patterns on Ring Lattices, May, 2021

LEADERSHIP AND SERVICE

Leadership

- Fall 2021 **Brown Division of Applied Mathematics - Directed Reading Program**, Brown University
- 2019 - 2020 **Brown Applied Math Undergraduate/Graduate Mentoring Program**, Brown University

Service

- 2021 - 2023 **Brown SIAM Student Chapter Executive Board**, Brown University
- 2021 - 2023 **Sheridan Center Departmental Graduate Student Liaison**, Brown University

TEACHING

Experience

- Fall 2025 **(Tentative) Introduction to Differential Equations with Linear Algebra**, Lecturer, University of Colorado Boulder
- Spring 2021 **Applied Ordinary Differential Equations**, Teaching Assistant, Brown University
- Fall 2020 **Operations Research: Deterministic Models**, Teaching Assistant, Brown University
- Fall 2018 **Single Variable Calculus**, Teaching Assistant, Dickinson College
- Fall 2017 **Introduction to Calculus**, Teaching Assistant, Dickinson College
- Fall 2016 **Single Variable Calculus**, Teaching Assistant, Dickinson College

Pedagogical Training

- Fall 2020 **Sheridan Center Certificate I: Reflective Teaching**, Brown University
Introductory seminar in a cross-disciplinary setting that helps develop and refine fundamental, evidence-based teaching skills and strategies